Assignment 3 - Math

TW Lab 2025

Write LaTeX code to output the following sections as it is. Use LaTeX reference wherever required. You may also explore the Math menu provided in the Menu Bar.

1 Basics

This famous Einstein's mass-energy equation, $E = m.c^2$ should be rendered inline. However the following equation of motion should be rendered in a block:

$$v = u + a.t$$

2 Fractions & Roots

$$x_i = \frac{\sqrt{a^2 + b_i^2}}{c_i}$$

3 Symbols & Math-fonts

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \quad \forall x \in \mathbf{R}\ \exists\ \varepsilon > 0 \Rightarrow |x| < \varepsilon$$

$$\mathcal{F}(\theta) = \sum_{k=0}^{n} \binom{n}{k} \theta^k (1-\theta)^{n-k}$$

4 Calculus

$$\frac{du}{dx} + \frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

$$\int_0^1 (3x^2 + 2x + 1)\, dx$$

5 Multi-line equations

5.1 Centering & alignment

$$x = 5 \tag{1}$$
$$x + y = 3 \tag{2}$$

$$x = 5 \tag{3}$$
$$x + y = 3 \tag{4}$$

5.2 Multiline equation

$$(x+y+z)^3 = (x+y+z).(x+y+z).(x+y+z) = x^3+y^3+z^3+ \\ 3x^2y+3xy^2+3y^2z+3yz^2+3x^2z+3xz^2+6xyz$$

6 Matrices

$$\begin{matrix} 1 & 2 \\ 3 & 4 \end{matrix} \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} \quad \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$